

Deterministic geometric Savitzky-Golay dynamics estimator

Shared parameter update: reference -> estimated

mass [kg]	2.351597591 -> 2.364645235	delta=+0.0130476
ratio=1.0055484		
CoG x [m]	-0.002024708562 -> -0.02352232972	delta=-0.0214976
ratio=-		
CoG y [m]	-3.052657894e-05 -> -0.02789511448	delta=-0.0278646
ratio=-		
CoG z [m]	0.009509749599 -> 0.04764633427	delta=+0.0381366
ratio=-		
rotor effectiveness 1	1 -> 1.035355554	delta=+0.0353556 ratio=1.0353556
rotor effectiveness 2	1 -> 0.9644484578	delta=-0.0355515 ratio=0.96444846
rotor effectiveness 3	1 -> 0.8893658912	delta=-0.110634 ratio=0.88936589
rotor effectiveness 4	1 -> 0.9515060466	delta=-0.048494 ratio=0.95150605
rotor command lag [s]	0.004997730255 -> 0.1900045394	delta=+0.185007
ratio=-		

Inertia matrix [kg m^2]

reference:

```
[ 0.06500006148, -7.27899253e-07, 1.901508003e-05]
[-7.27899253e-07, 0.06495265634, 5.9167305e-08]
[ 1.901508003e-05, 5.9167305e-08, 0.1289921107]
```

estimated:

```
[ 0.06512159438, -0.0007975215617, -0.0001475137112]
[-0.0007975215617, 0.06595091864, -0.000655335646]
[-0.0001475137112, -0.000655335646, 0.1287760408]
```

Principal moments: reference -> estimated

principal moment 1 [kg m^2]	0.06495264516 -> 0.06463397962	delta=-0.000318666
ratio=0.99509388		
principal moment 2 [kg m^2]	0.06500006701 -> 0.06643138978	delta=+0.00143132
ratio=1.0220203		
principal moment 3 [kg m^2]	0.1289921163 -> 0.1287831844	delta=-0.000208932
ratio=0.99838027		

Optimization

joint dynamics loss: 846.870736285
Gaussian-prior cost: 14.3972413257
selected strict-ZOH rotor lag [s]: 0.190004539397

Bag failure_1_single

reconstruction position RMSE [m]: 95.6506383427
reconstruction position component RMSE [m]: [8.738478 10.835155 94.632357]
reconstruction orientation RMSE [deg]: 151.587667962
gyro consistency RMSE xyz [rad/s]: [44.492807 42.22692 23.599488]
specific-force consistency RMSE xyz [m/s^2]: [51.878775 98.570744 44.815388]
inferred external force RMS xyz [N]: [9.693376 10.021167 9.022586]
inferred external torque RMS xyz [N m]: [4.634427 6.587026 9.522945]
bag dynamics loss: 846.870736285
gimbal command lag [s] 0.004998683929 -> 0.02149275253

Recorded command timestamp intervals (data-derived)

Bag failure_1_single

rotor: median dt=0.00499773026 s (200.091 Hz), min=2.00271606e-05 s, max=0.0138893127 s

gimbal: median dt=0.00499868393 s (200.053 Hz), min=4.529953e-06 s, max=2.48282003 s

Selected optimizer diagnostics

status=1 success=True nfev=6 njev=6
message='gtol' termination condition is satisfied.
reported optimality=1.9842485698973178e-07
initial cost=861.2680039625353
final cost=861.2679776106611
cost reduction=2.63518742258384e-05
coordinate step L2=0.0012366333467694983
final gradient inf-norm=1.9842485698973178e-07
gradient tolerance satisfied=True
final Jacobian rank=14
final Jacobian cond(nonzero)=429.11118502627755
max directional finite-difference Jacobian relative error=4.050542697931485e-12